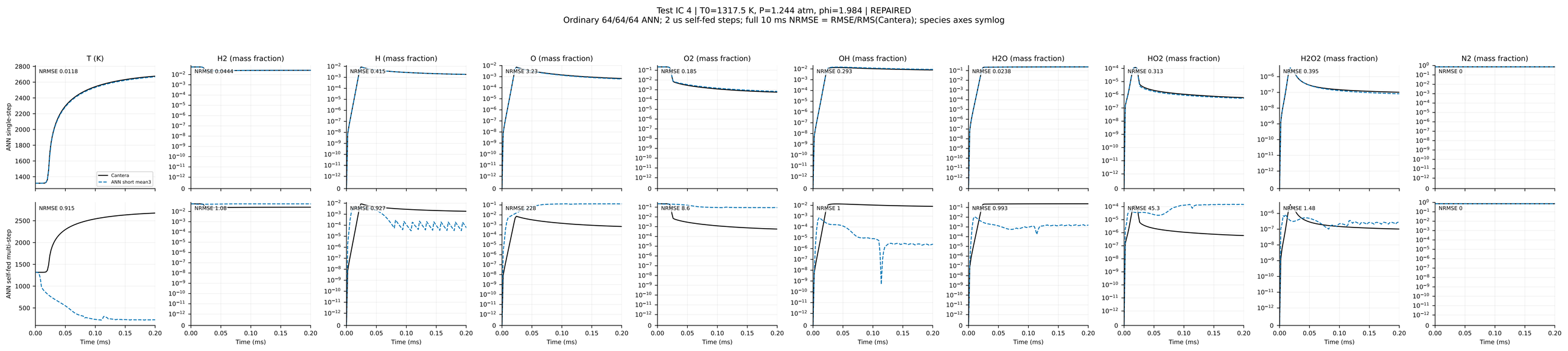
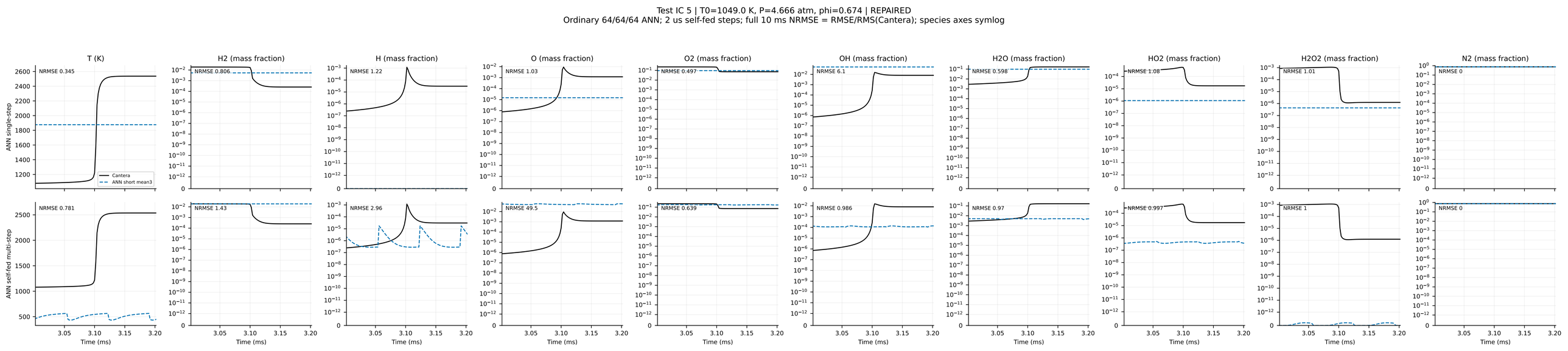
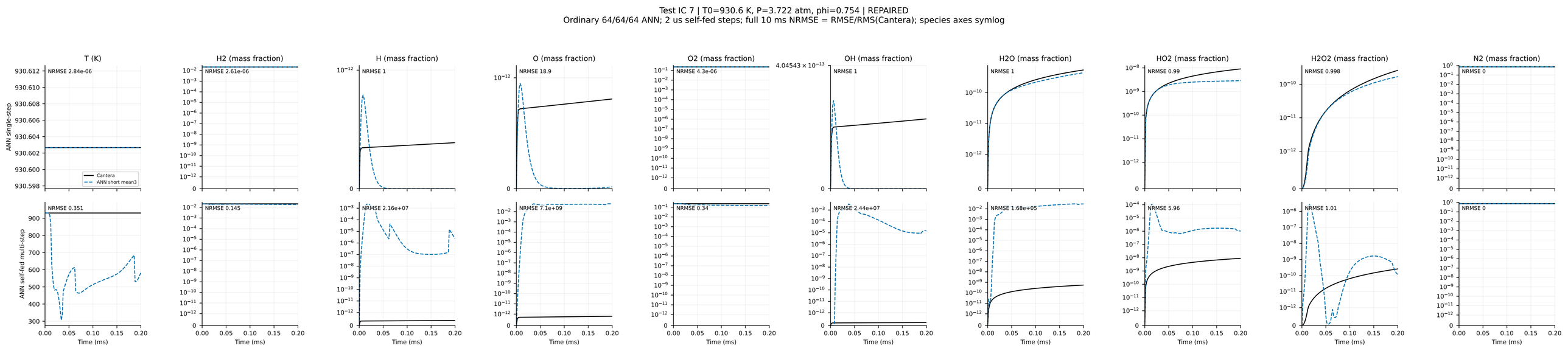
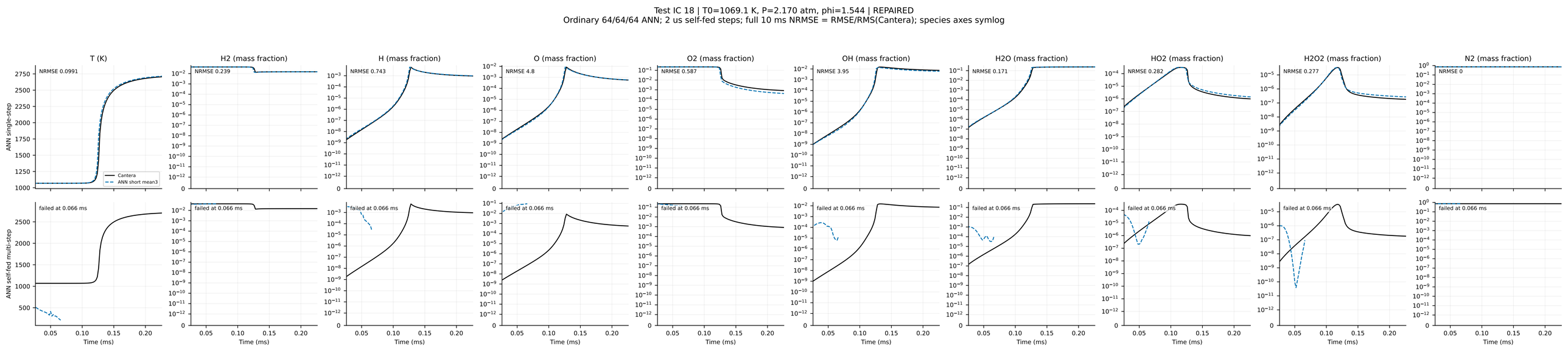
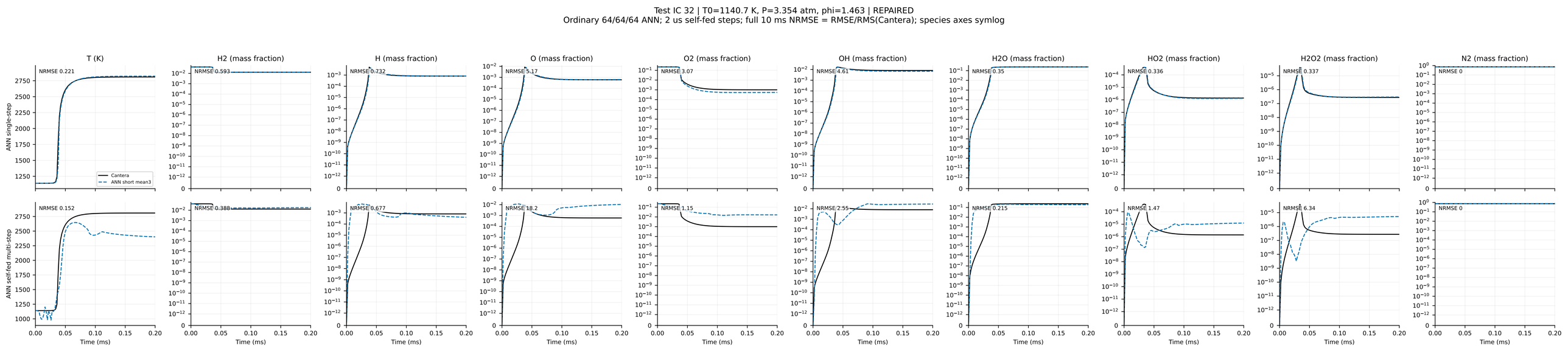
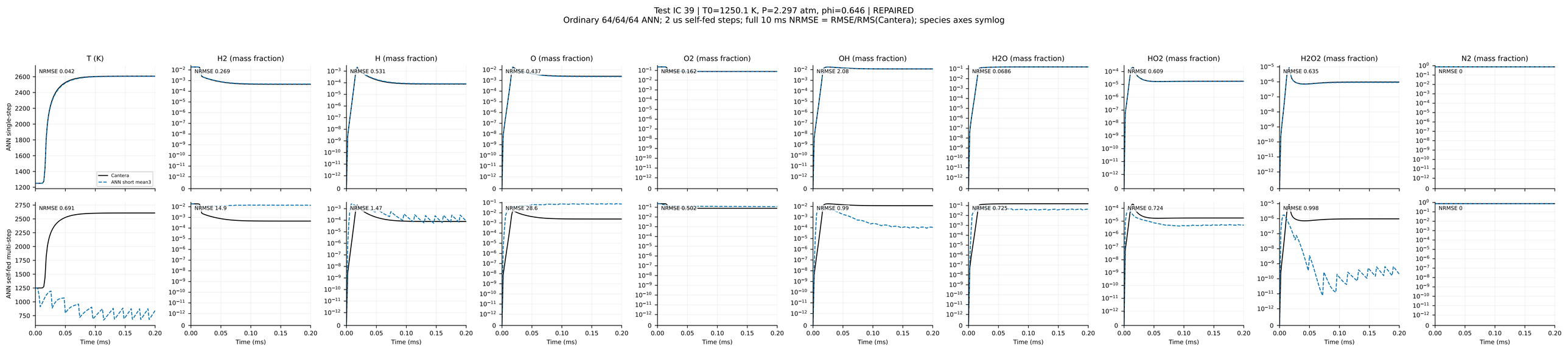

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0118
NRMSE 0.0444
NRMSE 0.415
NRMSE 3.23
NRMSE 0.185
NRMSE 0.293
NRMSE 0.0238
NRMSE 0.313
NRMSE 0.395
NRMSE 0
NRMSE 0.915
NRMSE 1.08
NRMSE 0.927
NRMSE 228
NRMSE 8.6
NRMSE 1
NRMSE 0.993
NRMSE 45.3
NRMSE 1.48
NRMSE 0
Cantera
ANN short mean3
Time (ms)

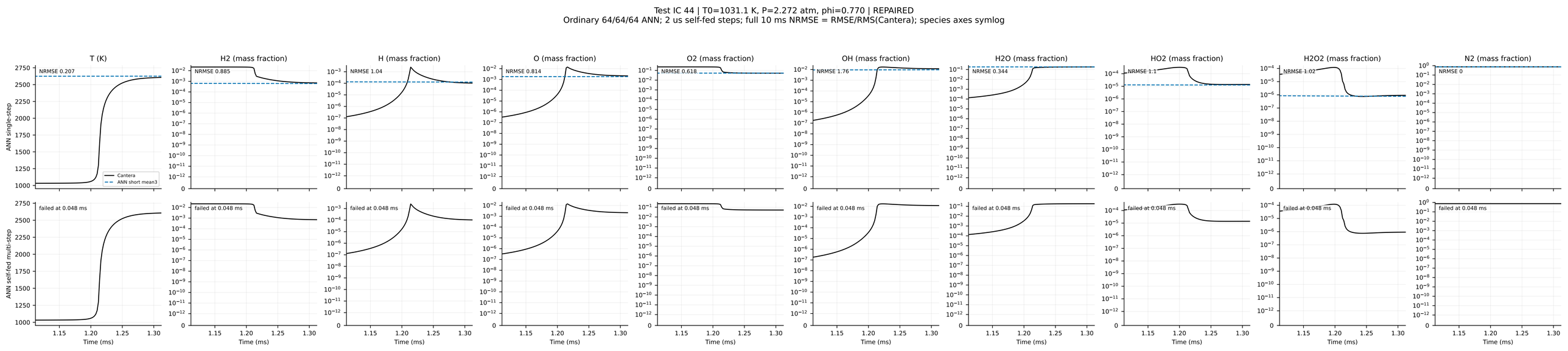

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.345
NRMSE 0.806
NRMSE 1.22
NRMSE 1.03
NRMSE 0.497
NRMSE 6.1
NRMSE 0.598
NRMSE 1.08
NRMSE 1.01
NRMSE 0
NRMSE 0.781
NRMSE 1.43
NRMSE 2.96
NRMSE 49.5
NRMSE 0.639
NRMSE 0.986
NRMSE 0.97
NRMSE 0.997
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

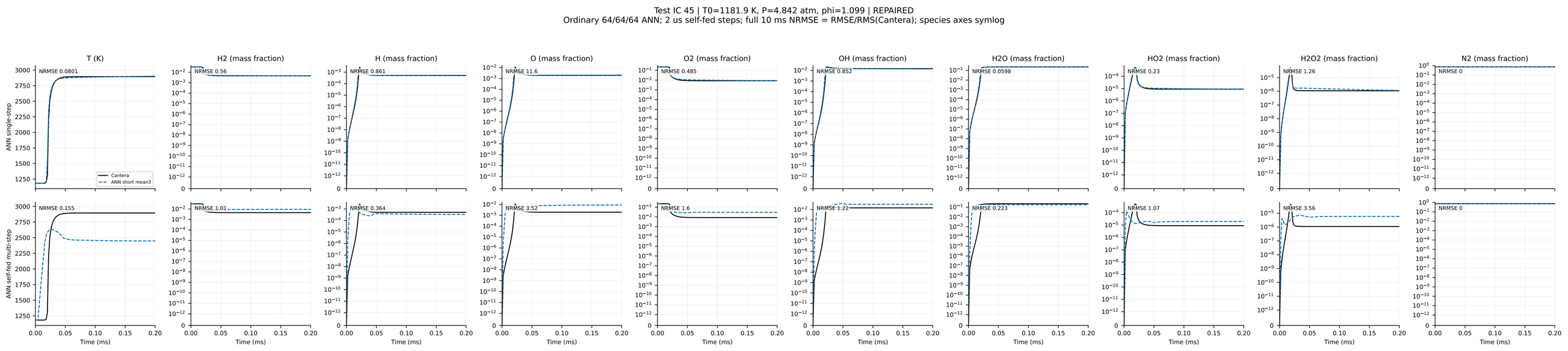

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 2.84e-06
NRMSE 2.61e-06
NRMSE 1
NRMSE 18.9
NRMSE 4.3e-06
NRMSE 1
NRMSE 1
NRMSE 0.99
NRMSE 0.998
NRMSE 0
NRMSE 0.351
NRMSE 0.145
NRMSE 2.16e+07
NRMSE 7.1e+09
NRMSE 0.34
NRMSE 2.44e+07
NRMSE 1.68e+05
NRMSE 5.96
NRMSE 1.01
NRMSE 0
Cantera
ANN short mean3
Time (ms)

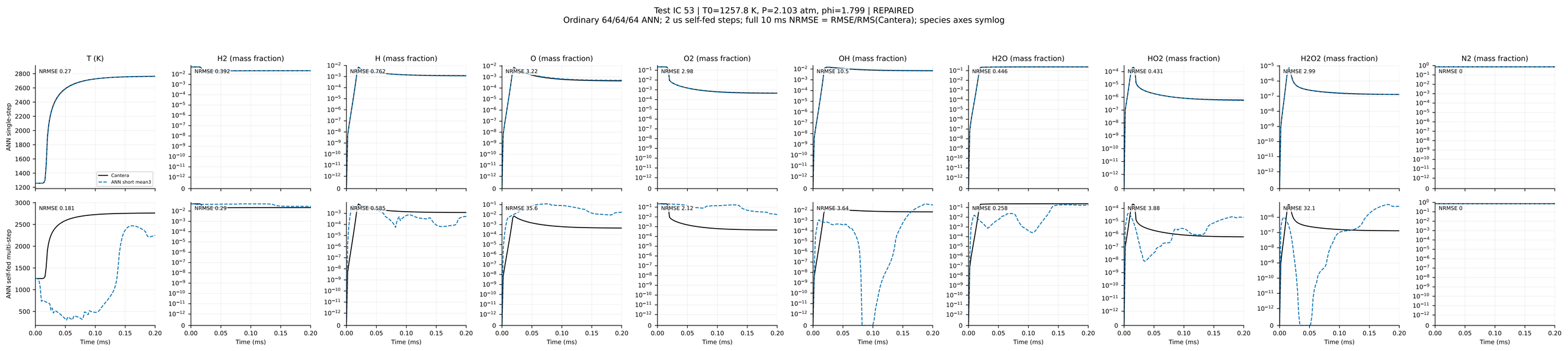

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
NRMSE 0.0991
NRMSE 0.239
NRMSE 0.743
NRMSE 4.8
NRMSE 0.587
NRMSE 3.95
NRMSE 0.171
NRMSE 0.282
NRMSE 0.277
NRMSE 0
Cantera
ANN short mean3
ANN single-step
ANN self-fed multi-step
failed at 0.066 ms
Time (ms)

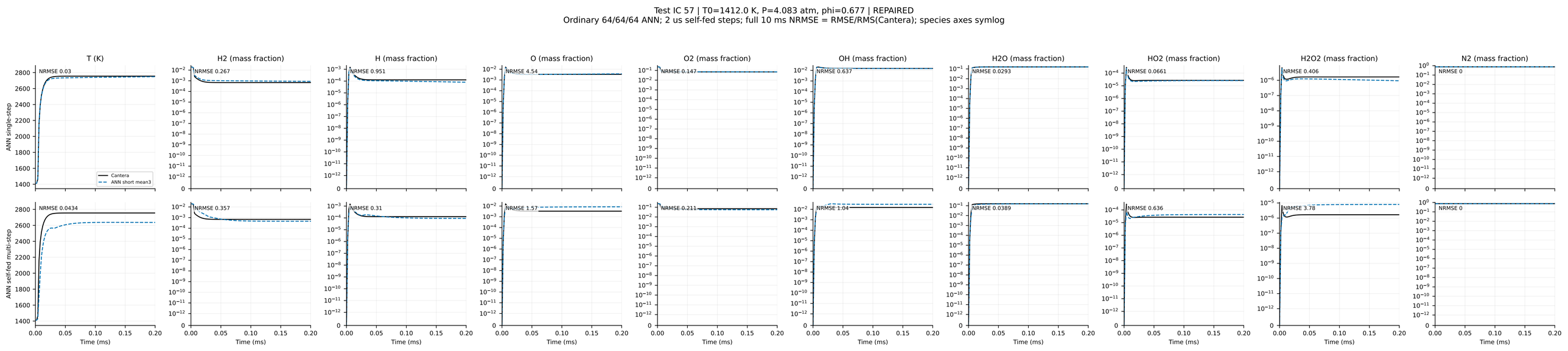

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.221
NRMSE 0.593
NRMSE 0.732
NRMSE 5.17
NRMSE 3.07
NRMSE 4.61
NRMSE 0.35
NRMSE 0.336
NRMSE 0.337
NRMSE 0
NRMSE 0.152
NRMSE 0.388
NRMSE 0.677
NRMSE 18.2
NRMSE 1.15
NRMSE 2.55
NRMSE 0.215
NRMSE 1.47
NRMSE 6.34
NRMSE 0
Cantera
ANN short mean3
Time (ms)

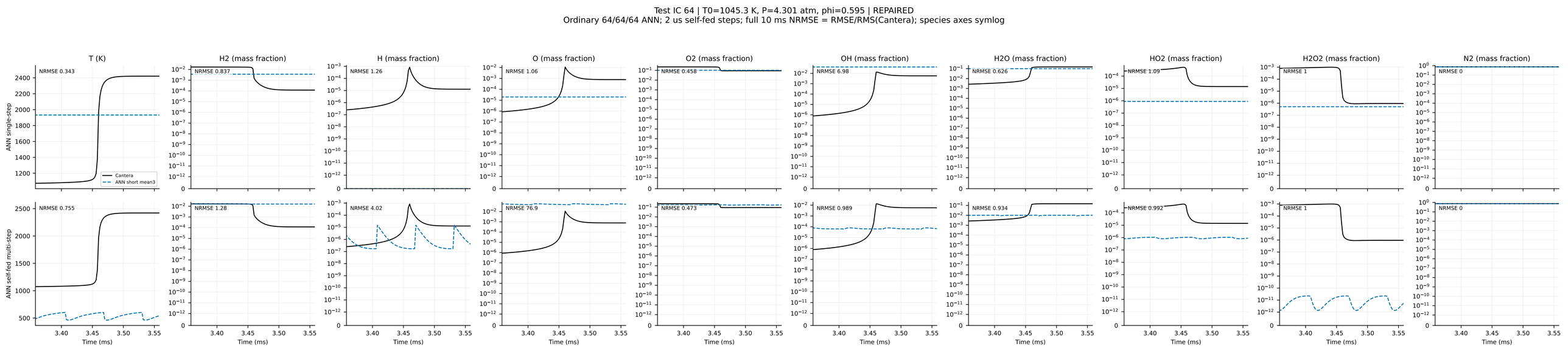

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.042
NRMSE 0.269
NRMSE 0.531
NRMSE 0.437
NRMSE 0.162
NRMSE 2.08
NRMSE 0.0686
NRMSE 0.609
NRMSE 0.635
NRMSE 0
NRMSE 0.691
NRMSE 14.9
NRMSE 1.47
NRMSE 28.6
NRMSE 0.502
NRMSE 0.99
NRMSE 0.725
NRMSE 0.724
NRMSE 0.998
NRMSE 0
Cantera
ANN short mean3
Time (ms)

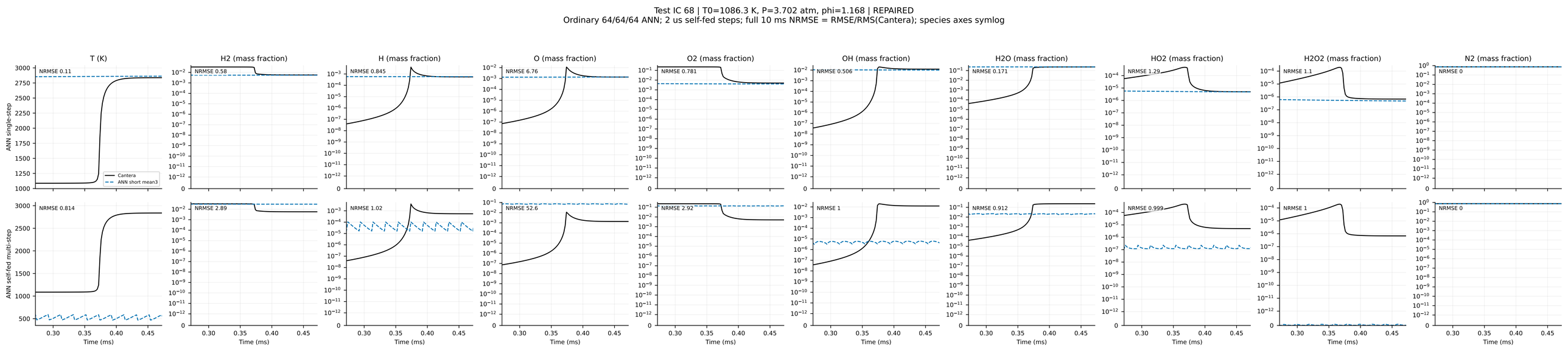

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.207
NRMSE 0.885
NRMSE 1.04
NRMSE 0.814
NRMSE 0.618
NRMSE 1.76
NRMSE 0.344
NRMSE 1.1
NRMSE 1.02
NRMSE 0
Cantera
ANN short mean3
failed at 0.048 ms
Time (ms)

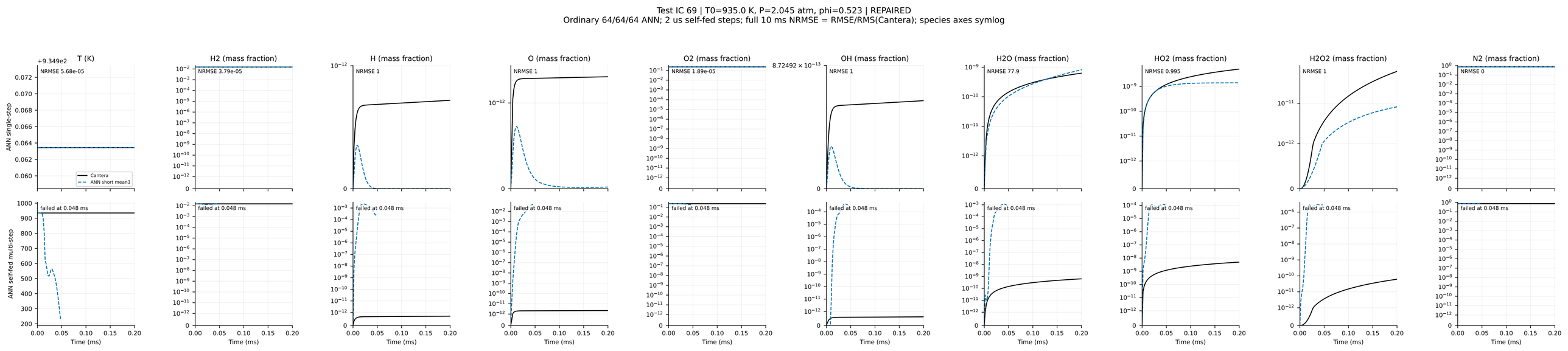

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0801
NRMSE 0.56
NRMSE 0.861
NRMSE 11.6
NRMSE 0.485
NRMSE 0.852
NRMSE 0.0598
NRMSE 0.23
NRMSE 1.26
NRMSE 0
NRMSE 0.155
NRMSE 1.01
NRMSE 0.364
NRMSE 3.52
NRMSE 1.6
NRMSE 1.22
NRMSE 0.223
NRMSE 1.07
NRMSE 3.56
NRMSE 0
Cantera
ANN short mean3
Time (ms)

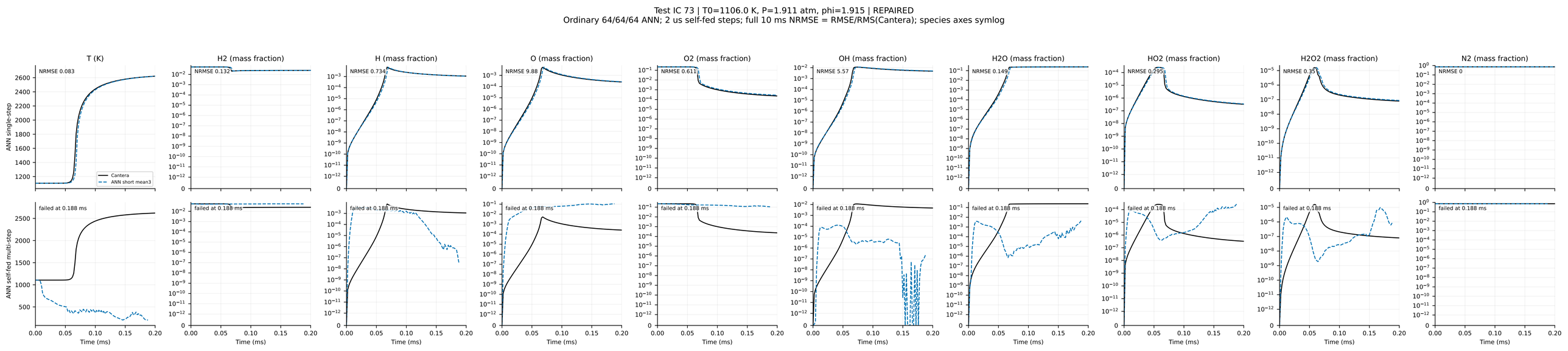

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.27
NRMSE 0.392
NRMSE 0.762
NRMSE 3.22
NRMSE 2.98
NRMSE 10.5
NRMSE 0.446
NRMSE 0.431
NRMSE 2.99
NRMSE 0
NRMSE 0.181
NRMSE 0.29
NRMSE 0.585
NRMSE 35.6
NRMSE 2.12
NRMSE 3.64
NRMSE 0.258
NRMSE 3.88
NRMSE 32.1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

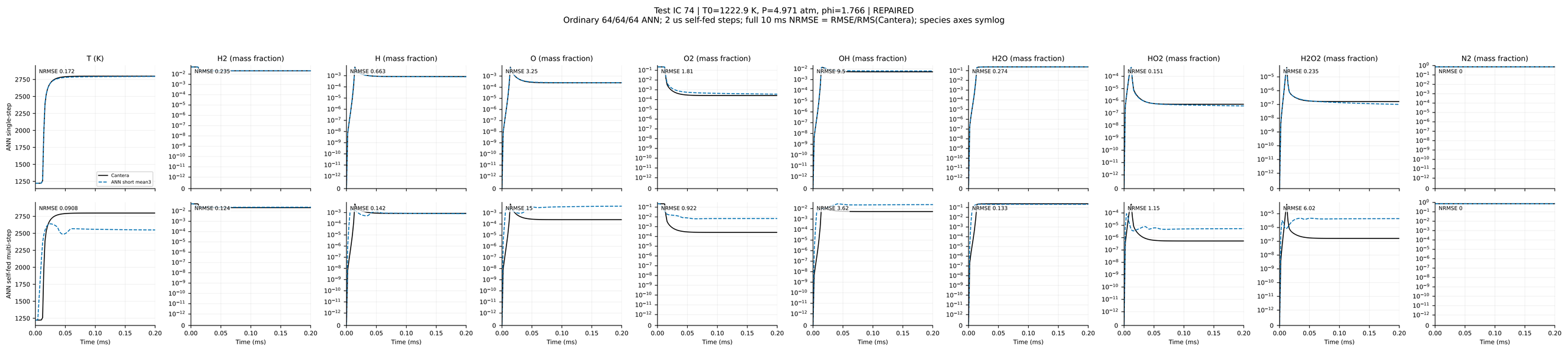

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.03
NRMSE 0.267
NRMSE 0.951
NRMSE 4.54
NRMSE 0.147
NRMSE 0.637
NRMSE 0.0293
NRMSE 0.0661
NRMSE 0.406
NRMSE 0
NRMSE 0.0434
NRMSE 0.357
NRMSE 0.31
NRMSE 1.57
NRMSE 0.211
NRMSE 1.04
NRMSE 0.0389
NRMSE 0.636
NRMSE 3.78
NRMSE 0
Cantera
ANN short mean3
Time (ms)

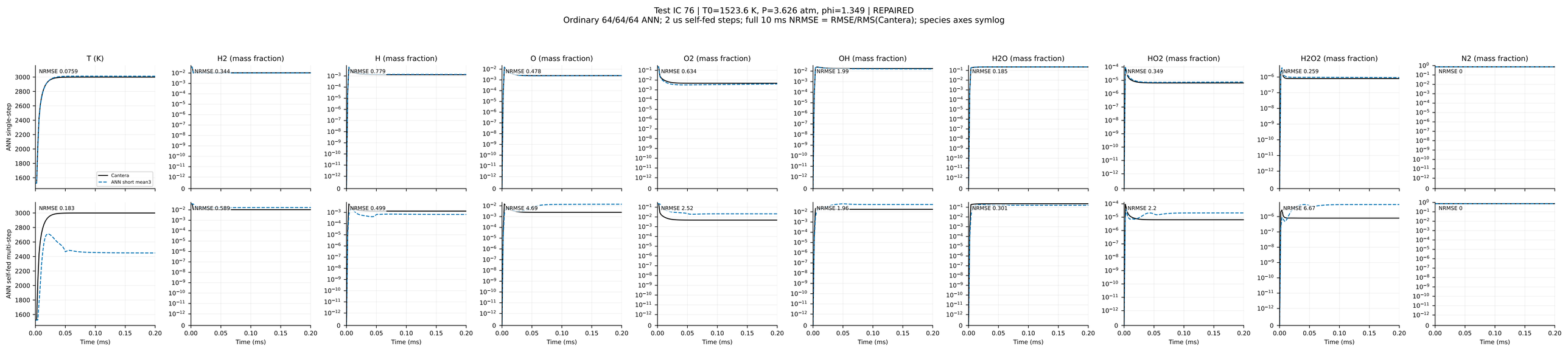

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.343
NRMSE 0.837
NRMSE 1.26
NRMSE 1.06
NRMSE 0.458
NRMSE 6.98
NRMSE 0.626
NRMSE 1.09
NRMSE 1
NRMSE 0
NRMSE 0.755
NRMSE 1.28
NRMSE 4.02
NRMSE 76.9
NRMSE 0.473
NRMSE 0.989
NRMSE 0.934
NRMSE 0.992
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

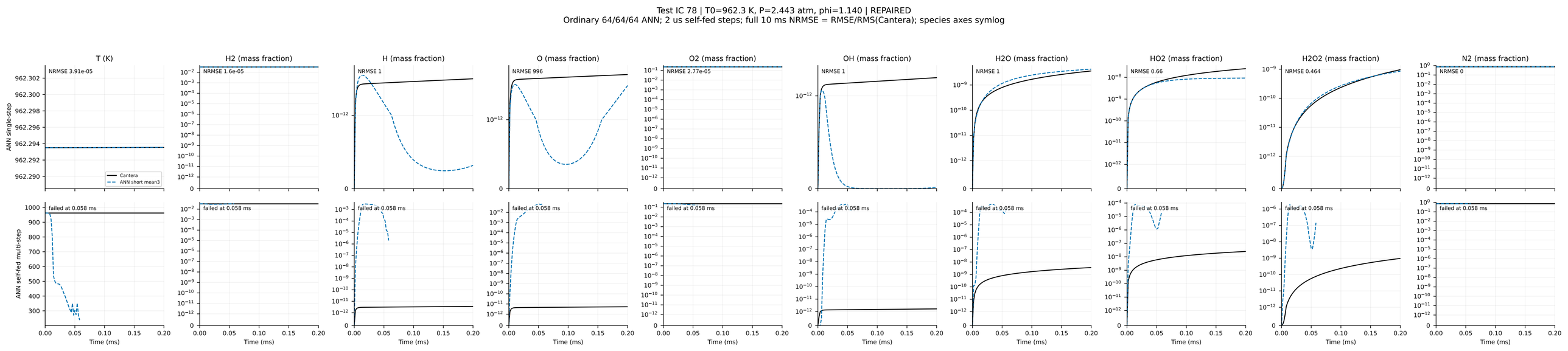

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.11
NRMSE 0.58
NRMSE 0.845
NRMSE 6.76
NRMSE 0.781
NRMSE 0.506
NRMSE 0.171
NRMSE 1.29
NRMSE 1.1
NRMSE 0
NRMSE 0.814
NRMSE 2.89
NRMSE 1.02
NRMSE 52.6
NRMSE 2.92
NRMSE 1
NRMSE 0.912
NRMSE 0.999
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
Time (ms)

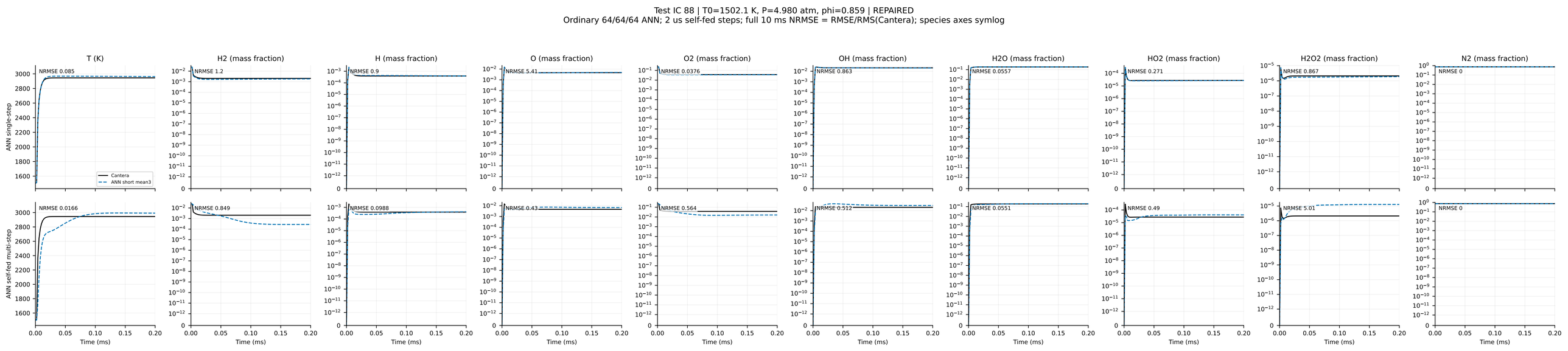

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
+9.349e2
NRMSE 5.68e-05
NRMSE 3.79e-05
NRMSE 1
NRMSE 1
NRMSE 1.89e-05
8.72492 × 10−13
NRMSE 1
NRMSE 77.9
NRMSE 0.995
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
failed at 0.048 ms
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.083
NRMSE 0.132
NRMSE 0.734
NRMSE 9.88
NRMSE 0.611
NRMSE 5.57
NRMSE 0.149
NRMSE 0.295
NRMSE 0.35
NRMSE 0
Cantera
ANN short mean3
failed at 0.188 ms
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.172
NRMSE 0.235
NRMSE 0.663
NRMSE 3.25
NRMSE 1.81
NRMSE 9.5
NRMSE 0.274
NRMSE 0.151
NRMSE 0.235
NRMSE 0
NRMSE 0.0908
NRMSE 0.124
NRMSE 0.142
NRMSE 15
NRMSE 0.922
NRMSE 3.62
NRMSE 0.133
NRMSE 1.15
NRMSE 6.02
NRMSE 0
Cantera
ANN short mean3
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0759
NRMSE 0.344
NRMSE 0.779
NRMSE 0.478
NRMSE 0.634
NRMSE 1.99
NRMSE 0.185
NRMSE 0.349
NRMSE 0.259
NRMSE 0
NRMSE 0.183
NRMSE 0.589
NRMSE 0.499
NRMSE 4.69
NRMSE 2.52
NRMSE 1.96
NRMSE 0.301
NRMSE 2.2
NRMSE 6.67
NRMSE 0
Cantera
ANN short mean3
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 3.91e-05
NRMSE 1.6e-05
NRMSE 1
NRMSE 996
NRMSE 2.77e-05
NRMSE 1
NRMSE 1
NRMSE 0.66
NRMSE 0.464
NRMSE 0
Cantera
ANN short mean3
failed at 0.058 ms
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.085
NRMSE 1.2
NRMSE 0.9
NRMSE 5.41
NRMSE 0.0376
NRMSE 0.863
NRMSE 0.0557
NRMSE 0.271
NRMSE 0.867
NRMSE 0
NRMSE 0.0166
NRMSE 0.849
NRMSE 0.0988
NRMSE 0.43
NRMSE 0.564
NRMSE 0.512
NRMSE 0.0551
NRMSE 0.49
NRMSE 5.01
NRMSE 0
Cantera
ANN short mean3
Time (ms)